



JWST NIRCам Reveals the Largest Known M dwarf Debris Disk Around TWA 10 and New Scattered-light Observations of the TWA 25 Debris Disk

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Abstract

We present JWST NIRCам observations of two M dwarf systems located in the TW Hydra association, TWA 10 and TWA 25. Both systems harbor detected debris disks in the F200W and F444W filters. Whereas the TWA 25 disk has been previously imaged, these observations represent the discovery and first images of the TWA 10 disk. In addition to planet searches within these systems, we also conduct an analysis of each debris disk, where the TWA 10 debris disk is characterized for the first time. We find that the TWA 10 debris disk is very large, with a radius of ~ 191 au, significantly greater than other known M dwarf debris disks. The TWA 25 disk hosts a sharp inner dust surface density power law and a moderate brightness asymmetry present at $2\ \mu\text{m}$, suggesting potential sculpting from inner planets and potentially enhanced collisional activity. Finally, we find one potential companion candidate within the TWA 10 system and two within the TWA 25 system, although the measured F200W – F444W color suggests that these candidates are likely background objects. Both systems do not have measured IR-excesses in their SEDs, where radiative-transfer modeling suggests that these disks (and potentially more M dwarf disks) were likely missed by previous disk detection surveys due to having low luminosity fractions.

Unified Astronomy Thesaurus concepts: Debris disks (363); Exoplanet systems (484); Near infrared astronomy (1093)

1. Introduction

Debris disks are optically thin, gas-poor, dusty disks found around $\sim 20\%$ of main-sequence stars (B. C. Matthews et al. 2014). They arise after the protoplanetary disk stage, i.e., after the majority of primordial gas and dust has dissipated, leaving behind formed exoplanets and planetesimals (M. C. Wyatt 2008, and references therein). Planetesimal belts, similar to the Asteroid and

Kuiper belts in our own solar system, can become dynamically excited or “stirred” through interactions with planets in the system (A. J. Mustill & M. C. Wyatt 2009), their own self-gravity (S. J. Kenyon & B. C. Bromley 2001), or other more rare mechanisms such as stellar flybys (S. Ida et al. 2000; S. J. Kenyon & B. C. Bromley 2002). This stirring leads to collisions between planetesimals, creating submicron- to millimeter-sized dust grains that can be observed through imaging at various wavelengths (M. C. Wyatt 2008; B. C. Matthews et al. 2014; A. M. Hughes et al. 2018).

Debris disks were first identified through infrared (IR) excess present in the stellar spectral energy distribution (SED),



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Table 1
Target Summary

Name	SpT	d (pc)	Age (Myr)	L_* ($L_\odot$)
TWA 10	M2V	57.5	9^{+2}_{-1}	0.106 ± 0.001
TWA 25	M0.5	53.6	9^{+2}_{-1}	0.246 ± 0.002

Note. Spectral types are taken from C. A. O. Torres et al. (2006) and G. J. Herczeg & L. A. Hillenbrand (2014) for TWA 10 and TWA 25, respectively. Distance is from Gaia DR3 (Gaia Collaboration et al. 2021) and system age is taken from N. Miret-Roig et al. (2025). The stellar luminosity is derived from the models fit to the photometry using the SED fitting code from B. Yelverton et al. (2019), which can be seen in Figure 5.

where the dust boosts the flux at mid-infrared (MIR) to far-infrared (FIR) wavelengths above what is expected for a stellar photosphere alone. Many surveys have been conducted to detect debris disks around nearby stars in the MIR–FIR with space-based instruments, including the Infrared Astronomical Satellite (IRAS), Wide-Field Infrared Survey Explorer (WISE), and the Herschel and Spitzer space telescopes. Observations with such instruments have been crucial in characterizing the SEDs of disk-hosting systems and determining the frequency of occurrence of debris disks around stars of various spectral types and ages (e.g., D. E. Trilling et al. 2008; R. I. Patel et al. 2014; B. Montesinos et al. 2016; B. Sibthorpe et al. 2018).

One issue with these detection surveys is the clear observational bias toward spectral type, such that there have been significantly less detections of IR-excesses around M dwarfs compared to AFGK stars. Additional ground-based imaging surveys targeted at M dwarfs have also yielded few detections (e.g., P. F. Cronin-Coltsmann et al. 2023). It remains unclear whether or not the lack of M dwarf disk detections is due to lower luminosity fractions (i.e., L_{disk}/L_*) (e.g., É. Morey & J.-F. Lestrade 2014) or potentially differing formation properties (e.g., lower initial disk masses E. Gaidos 2017). However, studies have found that M dwarf disks may be just as common as AFGK disks and have evaded detection due to observational limitations (P. Luppe et al. 2020). To date, only a small number of debris disks around M dwarfs have been detected over multiple observations, including well-known disks such as AU Mic (P. Kalas et al. 2004), TWA 7 (É. Choquet et al. 2016), and GSC 07396-00759 (E. Sissa et al. 2018). However, since the launch of the James Webb Space Telescope (JWST), at least one new M dwarf debris disk with no IR-excess detection has been discovered using the NIRCcam instrument (i.e., TWA 20 S. Palatnick et al. 2025). Thus, this discovery demonstrates that NIRCcam’s impressive sensitivity may provide the means to detect and spatially resolve faint M dwarf debris disks for the first time.

In this paper, we present debris disk observations around the TWA 10 and TWA 25 M dwarfs using the NIRCcam instrument on JWST. The properties of the systems can be found in Table 1. While the TWA 25 debris disk has been previously detected in scattered light with instruments on the Hubble Space Telescope (HST; É. Choquet et al. 2016; B. B. Ren et al. 2023) and the Very Large Telescope (VLT; N. Engler et al. 2025), the TWA 10 debris disk is a new detection and is now the second M dwarf debris disk to be discovered with JWST. Similarly to TWA 20, both the TWA

10 and 25 systems have no clear IR-excesses observed in their SEDs; these two systems have nondetections from Spitzer and Herschel. Here, we study the morphologies of these disks in our 2 μm and 4 μm observations, characterizing their geometrical, surface density, and scattering phase function (SPF) properties. Additionally, we search for potential planet candidates in both systems, as well as predict the maximum semimajor axis and minimum planet mass for each system assuming a singular massive planet sculpting the disk inner edge.

2. Data Observations and Reductions

2.1. Observations

Both targets were observed on June 16th of 2024 with JWST NIRCcam in coronagraphy mode (J. H. Girard et al. 2022) as part of the General Observers (GO) 4050 program: “Uncharted Worlds: Toward a Legacy of Direct Imaging of Sub-Jupiter Mass Exoplanets” (PI: A. Carter; A. Carter et al. 2023). The goal of the program is to search for new planetary companions around nearby stars using NIRCcam’s high sensitivity. Each system was imaged simultaneously in the following filters; F200W ($\lambda_{\text{pivot}} \sim 2 \mu\text{m}$, $\Delta\lambda \sim 0.46 \mu\text{m}$) and F444W ($\lambda_{\text{pivot}} \sim 4.4 \mu\text{m}$, $\Delta\lambda \sim 1.0 \mu\text{m}$). The F200W observations were taken with a field of view (FOV) of $10'' \times 10''$ and a pixel scale of 31 mas per pixel, while the F444W observations were taken with an FOV of $20'' \times 20''$ and a pixel scale of 63 mas per pixel. Both observations were taken with the MASK335R focal plane mask with an inner working angle of $0''.63$. Table 2 displays the summary of the data for TWA 10 and TWA 25. As part of the GO 4050 program, seven dedicated reference stars were also observed (in addition to the 22 targets observed) utilizing a 9-POINT-CIRCLE dither pattern.

2.2. Data Reduction

Both datasets were reduced using the codes `spaceKLIP` (J. Kammerer et al. 2022; A. L. Carter et al. 2023) and `Winnie`, which is a `spaceKLIP` compatible code designed for JWST coronagraphic disk observations (K. Lawson et al. 2023). The raw science and reference images are first reduced with the `spaceKLIP` reduction pipeline in three stages. In stage 1, the pipeline first applies detector-level corrections such as saturation, superbias, nonlinearity, and dark current, followed by ramp fitting of the raw, uncalibrated data. During this stage, we turn off the $1/f$ noise correction as removing this noise can lead to artifacts in images with disks. The data are then further processed in stage 2 of the pipeline, where additional correction and calibration are performed, including median background subtraction and identifying bad pixels. In the final stage, the data are bad-pixel repaired, recentered to ensure that all reference and science images are aligned, and padded with an 80 pixel border of nans to avoid any images being cut off when rotated to north up.

Once the data are reduced, `Winnie` is used to perform reference-star differential imaging (RDI) to produce a single PSF-subtracted image of the target. Similar to previous studies of debris disks observed with NIRCcam (see K. A. Crotts et al. 2025; S. Palatnick et al. 2025), we use the entire GO 4050 library for our reference observations to improve our PSF subtraction. We additionally include the other target stars, minus those with high levels of noise or debris disks, as we

Table 2
Summary of the Data Used in This Paper

Name	Filter	Date	Readout Pattern	$N_{\text{ints}}/N_{\text{groups}}/N_{\text{frames}}$	t_{int} (s)	t_{exp} (s)
TWA 10	F200W/F444W	2024 Jun 16	DEEP8	5/18/8	372	1865
TWA 25	F200W/F444W	2024 Jun 16	DEEP8	5/18/8	372	1865

Note. N_{ints} is the total number of integrations, N_{groups} is the number of groups per integration, N_{frames} is the number of frames per group, t_{int} is the effective integration time, and t_{exp} is the total effective exposure time.

find that this significantly improves the signal-to-noise (SNR) of our final reductions. The reference stars used (all M dwarfs but one) are the following; TWA 23 A, TWA 26, TWA 29, TWA 30 A, TWA 33, TWA 35, TWA 36, TWA 43, TWA 45, TWA 46, TWA 47, CD-23 9765, HD 89063, HD 101581, IRAS 09432-4847, IRAS 13041-5653, J1058-2346, i-Vel, and V Y Cr.

Finally, we perform model-constrained RDI (MCRDI; K. Lawson et al. 2022), a process that involves forward modeling and using best-fit synthetic disk models to fine-tune RDI constraints. The reasoning for using MCRDI is to optimize our PSF-subtracted images and minimize the loss of disk signal. For examples and further discussion of RDI versus MCRDI NIRCcam reductions, we point the interested reader to K. Lawson et al. (2023). The forward modeling process and results will be further discussed in Section 3. Our final MCRDI images of TWA 10 and TWA 25 can be seen in Figure 1.

3. Disk Forward Modeling

3.1. Procedure

In the previous section, we summarize the data reduction and PSF-subtraction process, where MCRDI is utilized to create the final images for both disks. Here, we further discuss the MCRDI process, which uses forward modeling to optimize the PSF-subtracted image while simultaneously finding a best-fit model to the data. We use a simple ring-like disk morphology based on J. C. Augereau et al. (1999) for our model, which assumes a two-parameter Henyey–Greenstein (HG; L. G. Henyey & J. L. Greenstein 1941) SPF. The disk geometry and surface density of our models are defined with the following parameters: the fiducial disk radius (r_0), the ratio between the scale height and r_0 at r_0 (h_0), the inclination (θ_{inc}), the position angle (θ_{PA}), the radial density power-law exponent interior and exterior to r_0 (α_{in} and α_{out} , respectively), and the vertical density exponent (γ). Additionally, the SPFs of the disk models are defined by two HG asymmetry parameters (g_1 and g_2 , respectively) and the weight for the asymmetry parameter g_1 (w_1).

To find the best-fit model, we use a minimization routine via the code LMFIT.MINIMIZE (M. Newville et al. 2025). For TWA 25, we utilize previous best-fit models of the disk found in É. Choquet et al. (2016) to define the initial starting values for each parameter. In the case of TWA 10, because this is the first resolved detection of the disk, the initial geometrical parameters (e.g., r_0 , θ_{inc} , and θ_{PA}) are estimated by eye. We start the minimization routine by fitting the F200W data, which have stronger disk detections. Additionally, we use the “Powell” method (M. J. D. Powell 1964) in order to find the local minimum for each parameter in a computationally efficient manner. In each case, several hundred models are

produced within a range of priors for each parameter, where an optimized model is then returned. The minimization routine is then run again with the best-fit parameters from the “Powell” method, this time using the “emcee” method (D. Foreman-Mackey et al. 2013), which utilizes Markov Chain Monte Carlo (MCMC) to further explore the parameter space and calculate uncertainties for each parameter. For both disks, we run the MCMC for 5000 steps with 30 walkers and a 300 step burn-in, resulting in $\sim 100,000$ models evaluated. Finally, we perform forward modeling for the F444W observations using the results for the F200W. While the geometrical and surface density profiles should not significantly change between wavelengths, the SPF may differ. Therefore, to fit the F444W observations, we repeat the same procedure as before, keeping the geometrical and surface density parameters fixed and only allowing the SPF parameters (g_1 , g_2 , and w_1) to vary. However, we note that the SPF parameters are difficult to constrain, especially for the F444W, due to the low SNR of the observations and the removal of flux at the forward scattering peak in the case of TWA 25.

The results of the forward modeling or both the Powell and emcee methods can be found in Table 3, where each row shows the best-fit values found for each parameter with 1σ uncertainties for the emcee method. Furthermore, Figure 2 shows our best-fit emcee models (middle panels) for the TWA 10 and TWA 25 disks, respectively, in both filters compared to our MCRDI reductions (left panels). The residuals from subtracting the model from the data are shown in the right panels. The best-fit disk models will be further discussed in Section 5.

4. Disk Analysis

4.1. Surface Brightness Profiles

In addition to deriving the disk geometry, surface density, and SPF parameters for each disk, we also derive surface brightness profiles in the F200W where the disks are brightest. Because the forward modeling has shown that the SPF for both disks is very difficult to constrain, we focus on what we can empirically learn from the surface brightness as a function of projected stellar separation.

We first spatially binned each image into 2×2 bins in order to increase the surface brightness S/N. Additional smoothing is performed by using a Gaussian filter with $\sigma = 1$ pixel. We then trace the disk spine, i.e., the (x, y) positions of the peak surface brightness along the projected major axis of the disk. This is done by rotating the data by the derived PA, so that the disk’s major axis is horizontal in the image. We then measure the surface brightness along vertical slices perpendicular to the disk’s major axis at various separations from the star (x -position) and fit these profiles with a Gaussian model. The mean value of the best-fit Gaussian is used as the y -position.

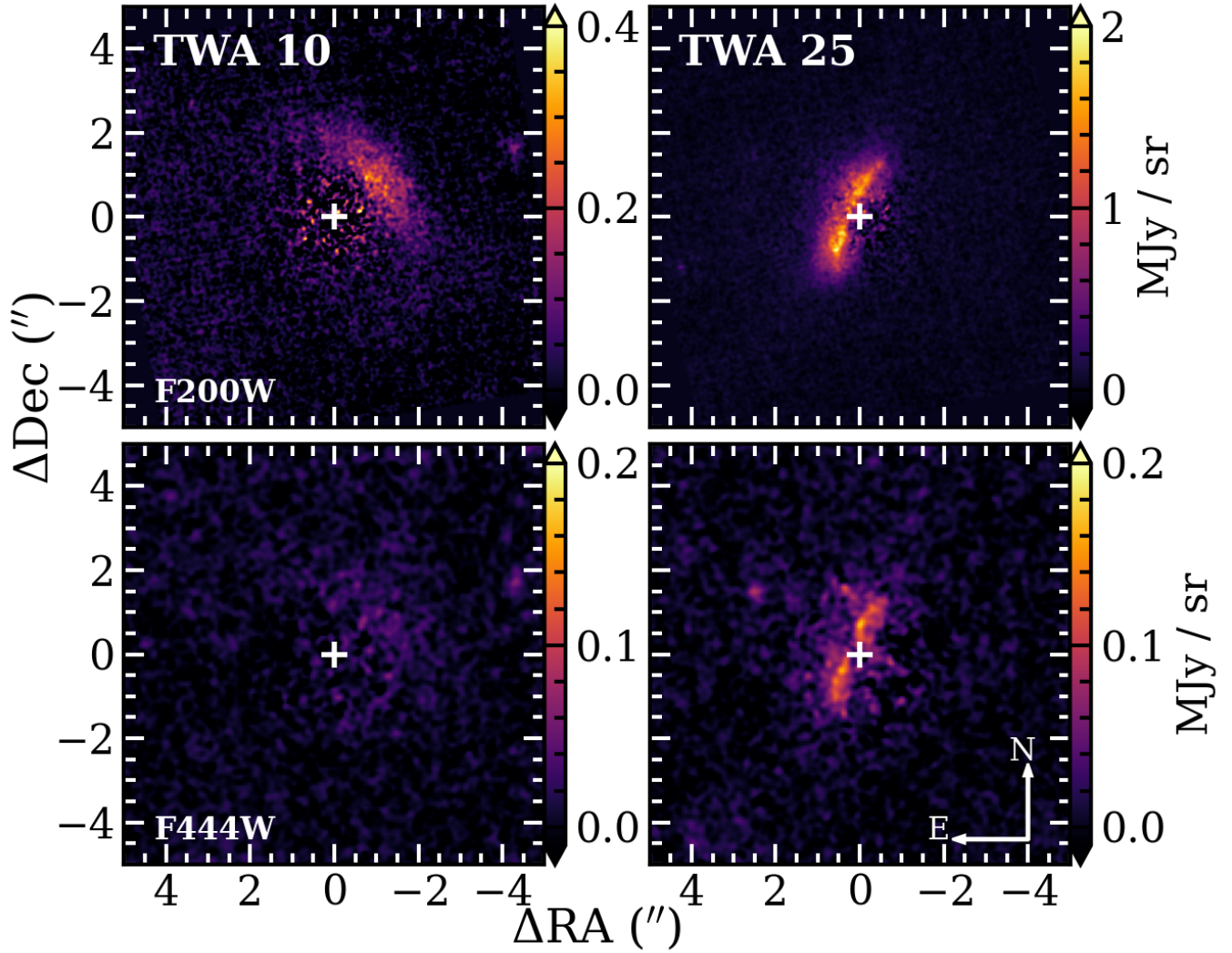


Figure 1. Final data reductions for (from left to right) TWA 10 and TWA 25 in the F200W (top) and F444W (bottom) filters. All images are rotated with north up, and the + represents the star location.

Next, we place a square 3×3 pixel aperture at each (x, y) position and take the average surface brightness in each aperture. The uncertainty is calculated by measuring the standard deviation within the same apertures placed on the model-subtracted residual maps. We split both surface brightness profiles between the east and west sides of the disk for comparison. The final surface brightness profiles are plotted in Figure 3.

The two disks present varying surface brightness profiles. In the case of TWA 10, the surface brightness profile peaks closest to the star and steadily decreases with increasing projected stellar separation, consistent with a typical debris disk SPF. However, the TWA 25 surface brightness profile differs, increasing between $0''$ and $\sim 0''.8$ from the star before decreasing past $1''$. This effect is systematic rather than physical. Given the smaller radius of the TWA 25 disk, the forward scattering peak within $\sim 0''.8$ becomes attenuated by the coronagraph. The shaded region in the bottom plot of Figure 3 shows where the coronagraphic throughput is $\lesssim 50\%$ for comparison.

Most importantly, we can use the surface brightness profiles to search for any asymmetries present. While, the TWA 10

appears fairly axisymmetric, the TWA 25 disk presents a significant brightness asymmetry. Mainly, the east side of the TWA 25 disk is moderately brighter than the west side of the disk between $0''.5$ and $1''.2$. A brightness asymmetry could be caused by an eccentric disk or potentially a dust density enhancement at these separations. The possibility of these scenarios will be discussed further in Section 5.2.

4.2. Disk Radius Comparison

Multiple studies of both scattered light and millimeter data have found tentative trends between debris disk radius and stellar luminosity, where the disk radius increases with stellar luminosity via a power law (e.g., L. Matrà et al. 2018, 2025; T. M. Esposito et al. 2020). Hence, debris disks around M dwarfs on average are found to be more compact than debris disks around hotter, more luminous stars. We therefore compare the TWA 10 and 25 disk radii to other spatially resolved debris disks to see if they fall within or outside these trends.

Figure 4 shows the radius versus the stellar luminosity for known debris disks. For comparison, we plot the measured radii and stellar luminosities for debris disks as part of the

Table 3
List of Best-fitting Model Parameters for Each Disk Using a Minimization Routine via the Powell and Emcee Methods

Name	Parameter	F200W Powell	F200W Emcee	F444W Powell	F444W Emcee
TWA 10	r_0 (au)	157.13	191.15 ± 6.53	...	...
	θ_{inc} (deg)	55.76	63.67 ± 3.36	...	...
	θ_{PA} (deg)	219.78	220.45 ± 3.67	...	...
	h_0	0.05 (fixed)	...	...	...
	α_{in}	4.78	4.53 ± 3.36	...	...
	α_{out}	-9.86	-8.45 ± 2.56	...	...
	γ	1.15	2.52 ± 1.29	...	...
	g_1	0.99	0.80 ± 0.13	0.99	0.02 ± 0.68
	g_2	0.99	0.57 ± 0.65	0.98	0.02 ± 0.68
	wg_1	0.55	0.80 ± 0.18	0.51	0.77 ± 0.17
TWA 25	r_0 (au)	62.42	63.81 ± 3.26	...	...
	θ_{inc} (deg)	78.09	77.91 ± 1.07	...	...
	θ_{PA} (deg)	337.04	337.07 ± 0.67	...	...
	h_0	0.05 (fixed)	...	...	...
	α_{in}	17.06	11.94 ± 5.37	...	...
	α_{out}	-3.83	-3.84 ± 0.31	...	...
	γ	1.79	2.76 ± 1.37	...	...
	g_1	0.97	0.90 ± 0.06	0.61	0.19 ± 0.65
	g_2	0.52	0.44 ± 0.19	0.61	0.10 ± 0.66
	wg_1	0.92	0.87 ± 0.07	0.99	0.75 ± 0.17

Note. Parameters that were fixed during the minimization process are labeled.

Gemini Planet Imager (GPI) sample (blue data points; T. M. Esposito et al. 2020). The solid black line and gray lines represent the power-law relationship derived from T. M. Esposito et al. (2020) using the GPI disk sample, while the dashed black line is the relationship derived from millimeter data via the REASONS survey for comparison (L. Matrà et al. 2025). We also highlight the other M dwarf disks that have measured radii (orange data points). This includes AU Mic and TWA 7, which are a part of the GPI sample, as well as the newly discovered TWA 20 debris disk also observed with NIRCcam (S. Palatnick et al. 2025) and the GSC 07396-00759 disk (E. Sissa et al. 2018). Additionally, we include two M dwarf disks that are not fully resolved with ALMA, GJ 2006 A, and AT Mic A (P. F. Cronin-Coltsmann et al. 2023), as upper limits on radius. The debris disk Fomalhaut C exists outside of the plot on the lower left ($L_* \sim 0.5 L_\odot$, $r_0 = 24.5$ au; K. Lawson et al. 2024).

While both the majority of resolved M dwarf disks fall in line with the derived relationships between radius and stellar luminosity, TWA 10 has a significantly larger radius. This discrepancy may have crucial implications for the stellar/disk and dynamical evolution of the TWA 10 system (discussed further in Section 5.1). However, the large range in debris disk radii around M dwarfs, which has become more apparent with the new disk discoveries, also suggests that the trend between disk radius and stellar luminosity does not hold up.

4.3. Disk SEDs

The detection of M dwarf debris disks has historically been difficult, leading to only a small handful of the brightest M dwarf disks being spatially resolved. However, the new discovery of M dwarf debris disks with NIRCcam, including TWA 10, may shed light on why this is the case. In Figure 5, we plot the stellar SEDs for TWA 10 and TWA 25. Grey data points are measured photometry consistent with the stellar SED model (blue line), which were fit using the SED fitting

software from B. Yelverton et al. (2019). The colored data points represent Spitzer/MIPS (orange) and Herschel/PACS (pink) photometry measurements. The photometry with citations is listed in Table 5, located in Appendix A. In both cases, the MIPS and PACS data are either 3σ upper limits or have large uncertainties consistent with the stellar SED within 1σ – 2σ . Although the TWA 25 disk was previously known, there has been no significant IR excess measured in its SED, similar to TWA 10.

Although we cannot robustly constrain the disk SEDs for these two systems without further observations of the disk emission, we can estimate the SED using the NIRCcam data and the limits set by MIPS and PACS. To calculate the disk SED, we utilize the radiative-transfer disk modeling code, MCFOST (C. Pinte et al. 2006). For the disk geometry and dust density parameters (R_{disk} , PA , inc , α_{in} , α_{out} , and γ), we use the best-fit values from Table 3 and similarly assume a scale height (h_0) of 0.05. For the dust grain parameters we assume astrosilicate grains with a grain size range of 0.1 – $1000 \mu\text{m}$ (the minimum dust grain size is small for M dwarfs) and a size distribution of -3.5 , which is expected for a collisional cascade (J. S. Dohnanyi 1969). Finally, the total dust mass is scaled to match the surface brightness of the F200W raw best-fit disk model. The resulting SEDs from the MCFOST models can be seen in Figure 5 as the purple lines.

Following the described process, we estimate a total dust mass of $\sim 6.7\text{e-}3 M_\oplus$ and $\sim 3.3\text{e-}3 M_\oplus$ for the TWA 10 and TWA 25 disks, respectively. These are typical dust masses for debris disks, albeit on the lower end for young moving group systems (e.g., L. Matrà et al. 2025). Additionally, the resulting disk SEDs are consistent with the limits measured with MIPS and PACS. While the TWA 25 disk SED sits only slightly below the PACS 3σ upper limits, peaking at a flux of 5.8 mJy, the TWA 10 disk SED is about an order of magnitude fainter than these upper limits, with a peak flux of 1.9 mJy. Dividing the disk SED sum from the stellar SED sum gives a luminosity fraction of $\sim 3.6\text{e-}5$ for TWA 10 and $\sim 6.9\text{e-}5$ for TWA 25. For

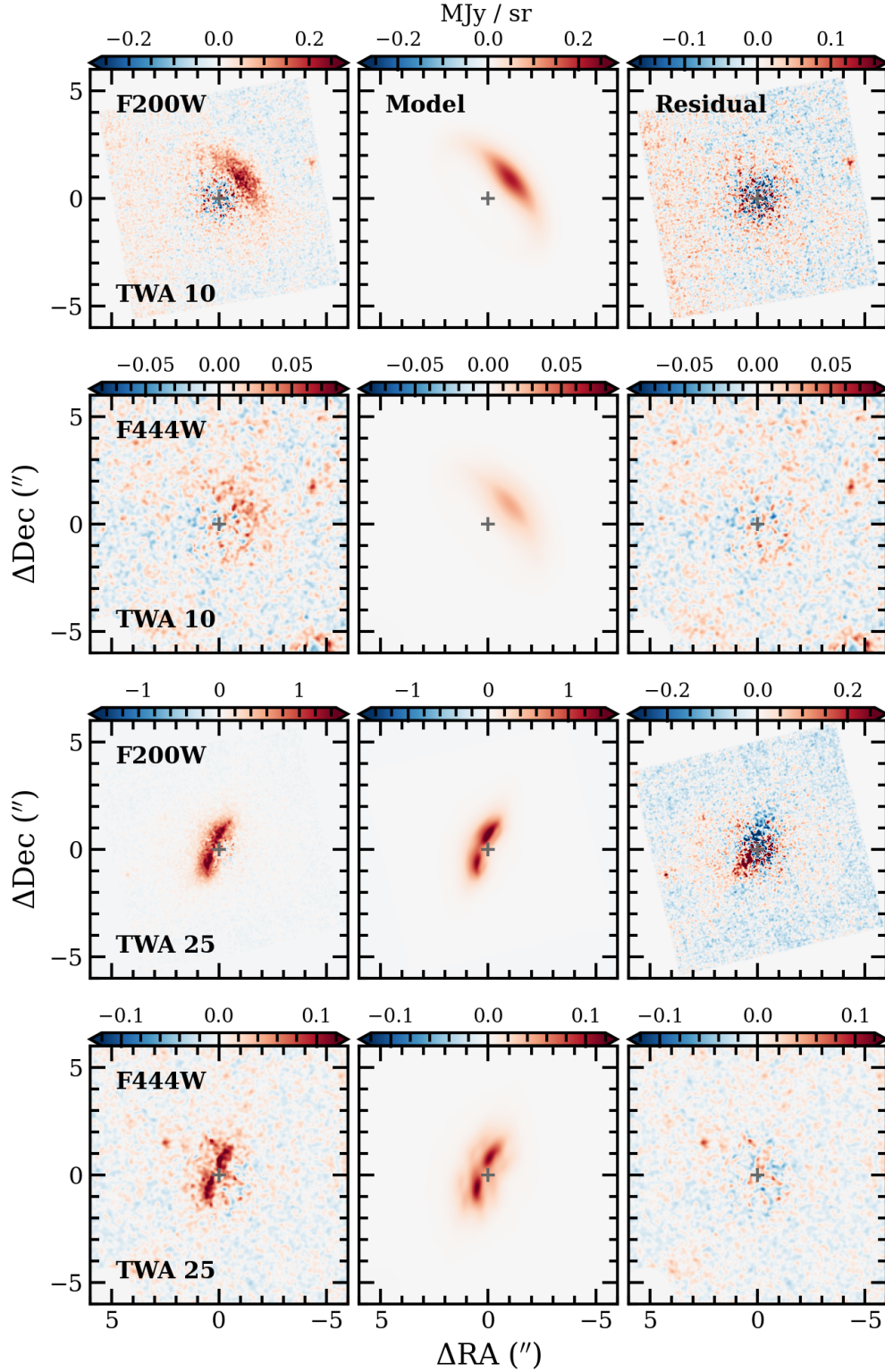


Figure 2. Forward modeling results for TWA 10 and TWA 25 in the F200W and F444W filters. The left panels are our MCRDI reductions, the middle panels are the best-fitting emcee forward models, and the right panels are the resulting residuals. All images are rotated with north up, and the + represents the star location.

context, cold debris disks ($T < 140$ K) with luminosity fractions of $< 10^{-4}$ have been difficult to detect in both scattered light (e.g., T. M. Esposito et al. 2020) and emission, particularly for M dwarfs (B. C. Matthews et al. 2014).

The main takeaway from this exercise is that the most likely reason the TWA 10 and TWA 25 debris disks (+ potentially

other M dwarf disks) were not detected with Spitzer and Herschel is because their disk emission is too faint. However, the disk SEDs remain highly uncertain given degeneracies between various dust grain properties and the total dust mass. Thus, future observations at longer wavelengths to detect the disk emission are necessary.

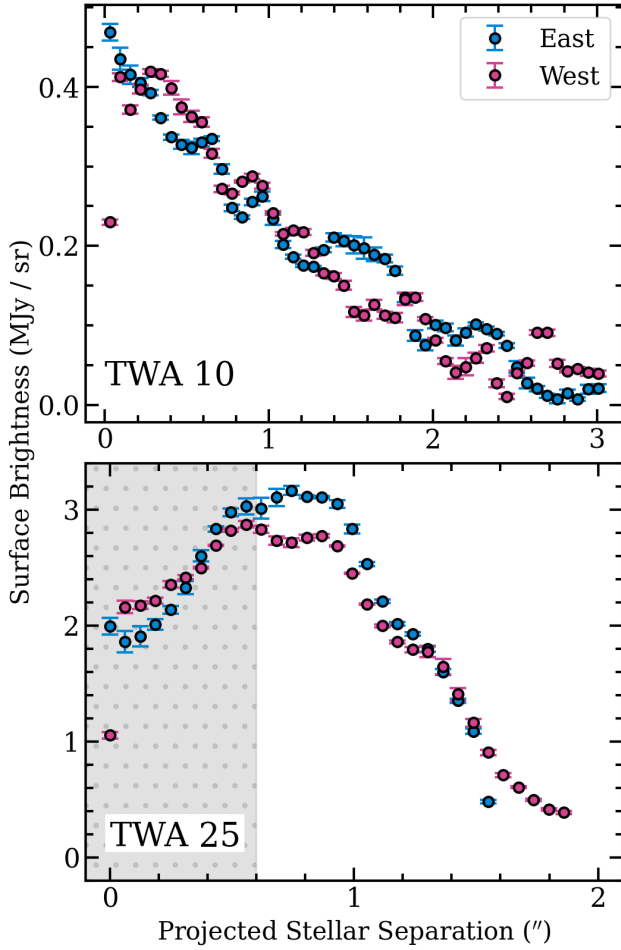


Figure 3. Surface brightness (SB) profiles for the TWA 10 (top) and TWA 25 (bottom) disks in the F200W as a function of projected separation from the host star. The blue data points represent the east side of the disk and the pink data points represent the west side of the disk. The gray shaded region represents the area where the throughput is $\lesssim 50\%$ ($\lesssim 0.6''$), as attenuation from the coronagraph affects the surface brightness of the TWA 25 disk at these separations.

5. Discussion

In the previous sections, we analyzed the TWA 10 and 25 debris disks in our NIRCcam observations by characterizing the disk geometry, surface density, SPF, and surface brightness. In the following sections, we provide further discussion of the implications of our results for both disks.

5.1. TWA 10

Our observations of TWA 10 provide the discovery and first detection of its debris disk. Previous studies have reported the lack of detected IR excess in the TWA 10 SED due to a debris disk (e.g., A. Schneider et al. 2012; P. Riviere-Marichalar et al. 2016). The SED can be seen in the top plot of Figure 5 in Appendix A, which shows no IR excess at wavelengths less than $30\ \mu\text{m}$ and was classified as nondetections by Spitzer/MIPS and Herschel/PACS at $70\text{--}500\ \mu\text{m}$. Additionally, the disk is relatively faint, with a surface brightness ranging between ~ 0.1 and $0.4\ \text{MJy sr}^{-1}$ or $\sim 0.002\text{--}0.008\ \text{mJy arcsec}^{-2}$ in F200W. Furthermore, the disk is barely detected in F444W, where significant spatial binning and smoothing is required to tease out the disk signal (see Figure 6). The lack of

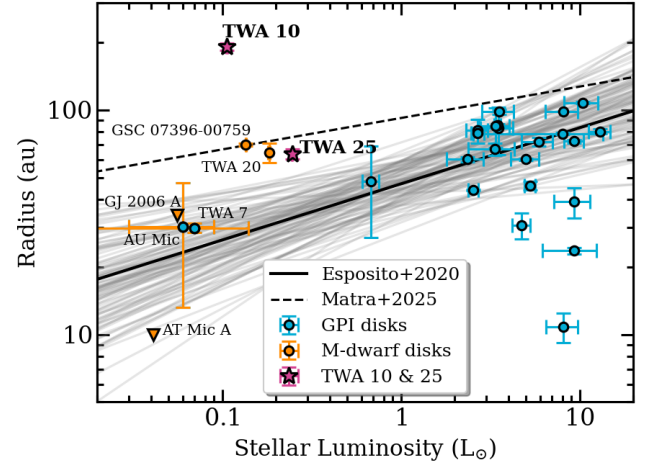


Figure 4. Disk radii (R_0) vs. stellar luminosity. The blue data points represent debris disks observed with GPI (T. M. Esposito et al. 2020), while the pink data points represent TWA 10 and 25. The orange data points highlight the other M dwarfs, where triangles are upper limits. The solid black and light gray lines represent the best-fit power law with 1σ uncertainties derived from T. M. Esposito et al. (2020) for the GPI sample. The dashed black line represents the best-fit power law derived from L. Matr  et al. (2025) for comparison.

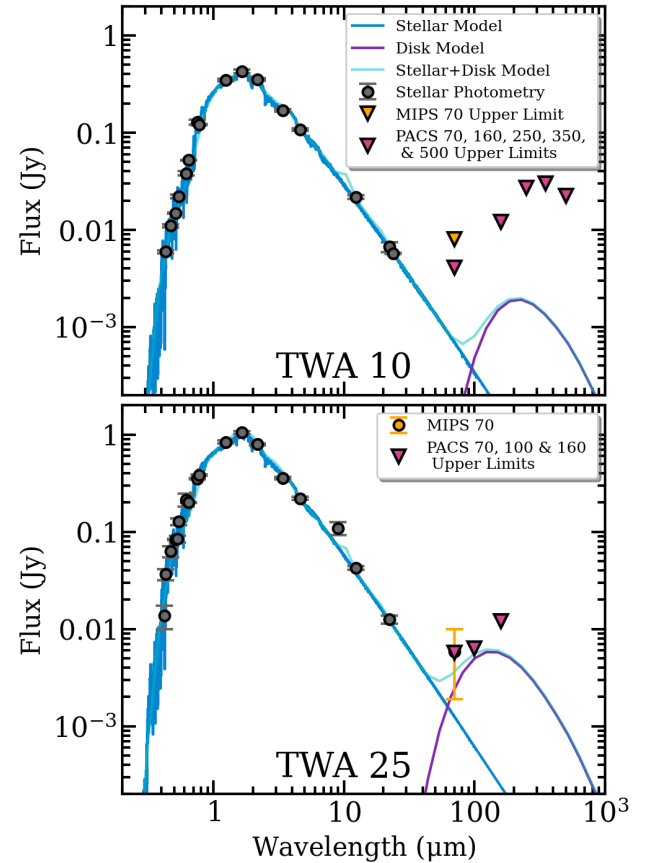


Figure 5. Stellar SEDs of TWA 10 (top) and TWA 25 (bottom). The gray data points are photometry measurements that are consistent with the stellar SED model (blue line), while the colored data points are measurements from Spitzer/MIPS and Herschel/PACS (triangles are 3σ upper limits). The estimated disk SED is shown by the purple line and the stellar+disk SED is represented by the light blue line.

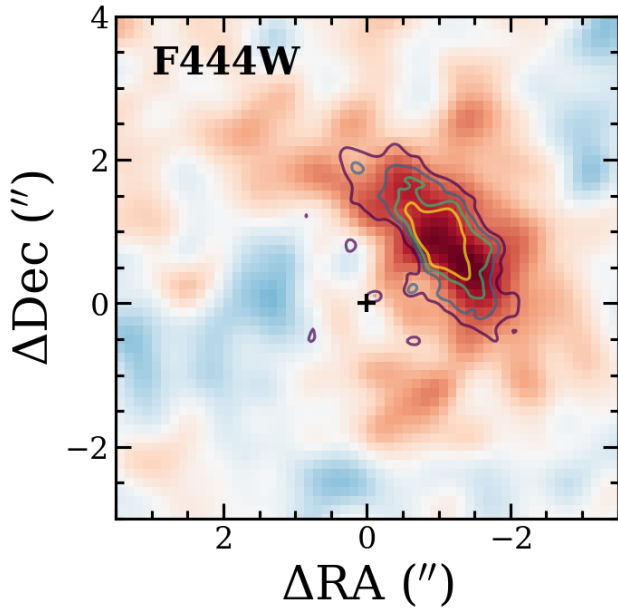


Figure 6. Close-in look of our convolved MCRDI reduction of TWA 10 in the F444W filter. The data is binned into 3×3 pixel bins and is further smoothed with a Gaussian with $\sigma = 2$ pixels. Contours represent the surface brightness of the disk in F200W (0.07, 0.10, 0.13, and 0.16 MJy sr $^{-1}$). While very faint, a clear signal is observed at the location of the disk as seen in the F200W.

detected IR excess, likely due to the faintness of the disk, explains why the TWA 10 disk was unknown until now; it is now the second M dwarf debris disk to be discovered by NIRCarn after TWA 20 (S. Palatnick et al. 2025). Our observations allow us to characterize the morphology of the TWA 10 debris disk for the first time.

Based on our modeling, we can learn basic information about the disk geometry. The disk is moderately inclined, with an inclination of $\sim 63^\circ.7$ and a PA of $\sim 220^\circ.5$ (measuring from east of north). We derive a disk radius of 191.15 ± 6.53 au via the emcee method, although the Powell method prefers a slightly smaller radius of ~ 157 au. In either case, this makes the TWA 10 disk the largest M dwarf debris disk to be imaged to date, with most M dwarf debris disks having radii $\lesssim 70$ au, as shown in Figure 4. In addition to the disk geometry, we find that it is difficult to strongly constrain the surface density profile, although results for α_{out} suggest a relatively steep surface density gradient in the outer edges of the disk. Finally, because only a small range of scattering angles are detected, we find that the SPF is also difficult to constrain, with no constraints made for g_2 in the F200W or any of the SPF parameters in the F444W. We note that the Powell method shows a preference for g_1 and g_2 parameters close to 1, which would suggest that the disk is highly forward scattering. However, higher SNR observations will be required to confirm these results.

Our most interesting result for the TWA 10 disk is its exceptionally large radius, as again, it is by far the largest M dwarf debris disk to be discovered to date. This discovery challenges our understanding of the formation and evolution of M dwarfs and their circumstellar environments. For example, observations of class II protoplanetary disks show similar correlations between disk size and spectral type, where disks around low-mass stars tend to be less massive and more compact (e.g., S. M. Andrews et al. 2013; N. van der Marel &

G. D. Mulders 2021; O. M. Guerra-Alvarado et al. 2025). While there are also more extended and structured protoplanetary disks observed around low-mass stars, many of these disks still have dust radii less than 100 au (e.g., N. T. Kurtovic et al. 2021; P. Pinilla 2022; O. M. Guerra-Alvarado et al. 2025). Thus, it is not clear if the disk originated at its current location (assuming the dust is tracing the planetesimal belt) or was moved out to a larger radius by another mechanism. The latter seems less likely, as radiation pressure (J. A. Burns et al. 1979) is negligible for M dwarfs due to their low luminosity. On the other hand, stellar winds, which are a dominant force for M dwarfs, can export dust both inwards and outwards (e.g., J. A. Burns et al. 1979; N. Pawellek et al. 2019). For example, both stellar wind forces have been used to explain the inner regions and extended outer halo for the AU Mic debris disk (e.g., L. E. Strubbe & E. I. Chiang 2006; C. Schüppler et al. 2015). However, in the case of AU Mic, the disk surface brightness profile peaks at ~ 30 – 40 au, i.e., at the relative location of the planetesimal belt (J.-C. Augereau & H. Beust 2006; L. E. Strubbe & E. I. Chiang 2006; M. A. MacGregor et al. 2013; K. Lawson et al. 2023). Because the TWA 10 disk surface brightness peaks at ~ 191 au, this suggests that the dust observed with NIRCarn is tracing the planetesimal belt. Future observations of the TWA 10 disk, especially at longer wavelengths, will be crucial for shedding more light on the disk’s origins by constraining the spatial distribution of the large grains, as well as the dust grain size distribution.

5.2. TWA 25

The TWA 25 debris disk is detected strongly in the F200W and faintly, but still significantly, in the F444W. These observations serve as the fourth detection of the disk (all of which are in scattered light), where other published observations of the disk include data taken with the NICMOS and STIS instruments on HST (É. Choquet et al. 2016; B. B. Ren et al. 2023) and most recently with the SPHERE instrument on the VLT (N. Engler et al. 2025). With NIRCarn’s improved sensitivity, we are able to better resolve the structure and confirm the morphology of the disk.

We compare our best-fit model values of the disk geometry with those derived from NICMOS and SPHERE. With our modeling, we are able to derive mostly consistent results with the geometrical parameters derived from the NICMOS and SPHERE modeling with additional constraints on the surface density profile. The consistent results include the disk inclination ($77^\circ.9 \pm 1^\circ.1$ compared to $75^\circ \pm 6^\circ$ and $78^\circ.3 \pm 1^\circ.5$ for NICMOS and SPHERE, respectively) and PA ($337^\circ.1 \pm 0^\circ.7$ compared to $336^\circ \pm 8^\circ$ and $336^\circ.8 \pm 1^\circ.8$). However, we derive a smaller radius than previous estimations (63.8 ± 3.3 au compared to 78 ± 17 au and 76 ± 2). Although the radius is still consistent with the NICMOS estimates within 1σ this is not the case for the SPHERE results. This discrepancy may be due to the sensitivity and data reduction techniques of the NICMOS and SPHERE observations. Whereas the NICMOS observations are simply low SNR, the SPHERE observations utilize techniques such as angular differential imaging (C. Marois et al. 2006), which can result in disk self-subtraction (evident by the difference in vertical width between observations) and thus skew the radius estimate. Additionally, the radius measurement from SPHERE is derived from fitting an ellipse to the processed image, which

does not take into account potential effects from the PSF-subtraction process. Although the backside of the TWA 25 disk is still not detected with NIRC*am*, the high sensitivity of the observations, lack of disk self-subtraction at the outer edges, and use of forward modeling suggest that our radius estimation are likely to be closer to reality. Finally, we find that the modeling prefers a steep inner dust density profile (α_{in}) of ~ 12 – 17 , although the uncertainty derived from the emcee method is large. If α_{in} is indeed steep, this may suggest that the disk is being sculpted by a planet within the disk’s inner radius.

For the disk SPF, our modeling prefers a high value for g_1 of $\gtrsim 0.9$, again suggesting a highly forward scattering disk at $2\ \mu\text{m}$ (we are unable to constrain the SPF in the F444W). However, because the flux from the disk is significantly removed at the location of the forward scattering peak, due to attenuation from the coronagraph, the results from this analysis should be taken with a grain of salt. For example, conducting the same forward modeling process (via the Powell method) with a single-component HG function leads to a lower preferred g value of 0.79 . Thus, future observations, where the flux is not significantly removed from the forward scattering peak, will be necessary to place tighter constraints on the SPF.

Additionally, a tentative brightness asymmetry appears to be present at $2\ \mu\text{m}$, where the SE side of the disk is moderately brighter than the NW side. This brightness asymmetry can also be seen in the residual profile after subtracting our best-fit model, where a significant positive residual is present in the SE side (see Figure 7). A similar brightness asymmetry also appears to be present in the NICMOS data, which is at a similar wavelength ($\lambda \sim 1.6\ \mu\text{m}$), although it is not clear if it is observed in the SPHERE data. We also confirm this brightness asymmetry between a separation of $\sim 0''.5$ and $1''.2$ in the surface brightness profile (see Figure 3). To further quantify the asymmetry, we place two equally sized rectangular apertures on either side of the disk between $0''.5$ and $1''.2$ in both the F200W and F444W images. We then take the sum and standard deviation of the flux in each aperture and divide the west summed flux from the east summed flux. Doing so, we find that the east side of the disk is roughly 1.10 ± 0.01 times brighter than the west side at $2\ \mu\text{m}$ while the $4\ \mu\text{m}$ data is consistent with being symmetric (1.02 ± 0.02). However, the actual surface brightness asymmetry may be more extreme than the initial measurements suggest due to artificial asymmetries introduced by the coronagraphic PSF core, which is oval shaped with skewed side lobes. This effect is evident from the best-fit model to the F200W data, where the coronagraphic PSF has caused the NW side of the disk to appear brighter than the SE side (although the same effect does not appear in the F444W model). Thus, we apply the same measurements to a version of our F200W reduction that is deconvolved from the instrumental PSF for comparison (shown in the top plot of Figure 7). As hypothesized, the asymmetry is even more extreme in the F200W deconvolved image, where we measure the east side of the disk to be 1.25 ± 0.04 times brighter than the west side.

Brightness asymmetries can occur through several mechanisms. For example, if the disk is eccentric, such as through sculpting from an eccentric planet, the disk pericenter will scatter more light from the star causing a pericenter glow (M. C. Wyatt et al. 1999). However, if this is the case, we

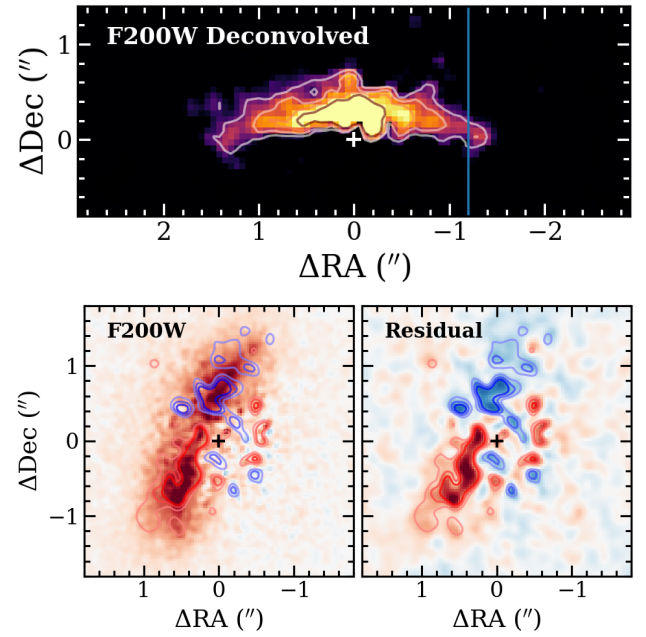


Figure 7. Top: our MCRDI reduction in the F200W filter deconvolved with the instrumental PSF. The image is rotated by the disk PA, so that the major axis is horizontal. Contours are added at 2, 5, 10, and 20 MJy sr $^{-1}$ to highlight the observed brightness asymmetry. Bottom left: our MCRDI reduction of TWA 25 in the F200W filter. Bottom right: the residual map after the best-fitting model of the TWA 25 debris disk is subtracted. The red contours represent positive residuals between 0.1 and 0.2 MJy sr $^{-1}$, while the blue contours represent negative residuals between -0.2 and -0.1 MJy sr $^{-1}$.

would expect to see a similar brightness asymmetry at both wavelengths. Additionally, we do not detect any significant offset of the disk from the star. One reason the disk may be asymmetric in brightness at different wavelengths is variations in dust grain properties across the disk, such as composition, porosity, or grain size. For example, this can be the result of a recent large collision between two rocky bodies in the system, a hypothesis that has been used to help explain the dust clumps and brightness asymmetries in other debris disks such as β Pic and HD 111520 (W. R. F. Dent et al. 2014; K. A. Crotts et al. 2022; J. W. Jones et al. 2023; I. Rebollido et al. 2024). However, the F200W and F444W observations should trace relatively the same dust grain population given the small difference in wavelength. Because the SNR is lower at $4\ \mu\text{m}$, it is possible that we simply do not have the sensitivity to detect the brightness asymmetry in the F444W filter. Future observations at longer wavelengths, such as with ALMA, would be useful to determine the spatial distribution of the larger grains, which can shed more light on the mechanisms shaping the TWA 25 disk.

6. Searching for Planets

In addition to analysis of the TWA 10 and 25 debris disks, we also search for potential planetary candidates in these systems that could be dynamically interacting with their disks. We are particularly interested in faint point sources within $\sim 5''$ as we can constrain their color between the F200W and F444W filters. We expect that candidates in the planetary regime are very red in color and, therefore, should appear in F444W images but not significantly in F200W images. Point sources seen outside of $5''$ may still be potential candidate

companions; however, it is less likely given their distances from the host star (higher likelihood of contamination from background objects) and we are unable to constrain their colors. Therefore, we do not consider them in this Section, although we constrain the location and F444W flux of these objects in Appendix B. We compute the contrast curves and planetary mass sensitivity of our observations and characterize several potential candidate companions found within the TWA 10 and 25 systems.

6.1. Contrast Curves

To calculate the 5σ contrast curves for our observations, we utilize `spaceKLIP` for which we input the disk model subtracted MCRDI reductions and the stellar SED. The resulting outputs are considered the raw 5σ contrast curves, where a second step is usually conducted in `spaceKLIP` to produce calibrated contrast curves, taking into account extra variables such as the throughput of its KLIP algorithm. However, because the data are reduced with MCRDI instead of KLIP, this step becomes unnecessary as there is no regime where the MCRDI throughput should impact point-source detectability at the 5σ level. Thus, the raw contrast curves, which measure the image noise and take into account coronagraphic throughput, are sufficient to evaluate the planet-mass sensitivity of our observations and the contrasts of any potential candidates. Our final contrast curves are plotted in Figure 8. We also include the contrasts for the three candidates identified and analyzed in Section 6.2.

We use our F444W 5σ contrast curves to also measure the planet-mass sensitivity of our observations. To do so, we use the standard ATMO-CEQ planet evolutionary models from M. W. Phillips et al. (2020) and the cloud-free BEX models with solar metallicity ($[M/H] = 0$) from E. F. Linder et al. (2019) for a system of age 10 Myr. However, we note that these models represent the most optimistic cases for the planet-mass sensitivity as clouds, higher metallicity, and nonequilibrium chemistry can suppress the planet flux (e.g., J. Mang et al. 2024; R. Bowens-Rubin et al. 2025; K. A. Crotts et al. 2025). The resulting planet-mass sensitivities are then inputted into `ExoDMC` (M. Bonavita 2020), a rendition of the MESS code (M. Bonavita et al. 2012) that calculates exoplanet detection maps by estimating the probability of detection of a synthetic planet population. We assume a Gaussian distribution for the eccentricity with a mean of 0 and a σ of 0.3. Additionally, we use the measured inclinations of the debris disks in both systems to add further constraints to the detection probability maps. This assumes that any planetary companions present are coplanar with the disks. Coplanar orbits are expected for exoplanets that formed within their protoplanetary disks and have been observed for other directly imaged exoplanets with debris disks (e.g., β Pic b: A.-M. Lagrange et al. 2010; B. Macintosh et al. 2014; E. L. Nielsen et al. 2014, HD 95086 b: J. Rameau et al. 2016, HR 8799 b-e: J. J. Wang et al. 2018). The resulting exoplanet detection probability maps can be found in Figure 9.

We compare our detection probability maps with the extent of the TWA 10 and 25 debris disks, as well as the expected maximum semimajor axis and minimum mass expected for a single planet actively sculpting the disk. The inner radius and extent of each disk is estimated using the best-fit r_0 , α_{in} , and α_{out} values and are represented by the black dashed lines and orange shaded regions in Figure 9. Based on the disk inner

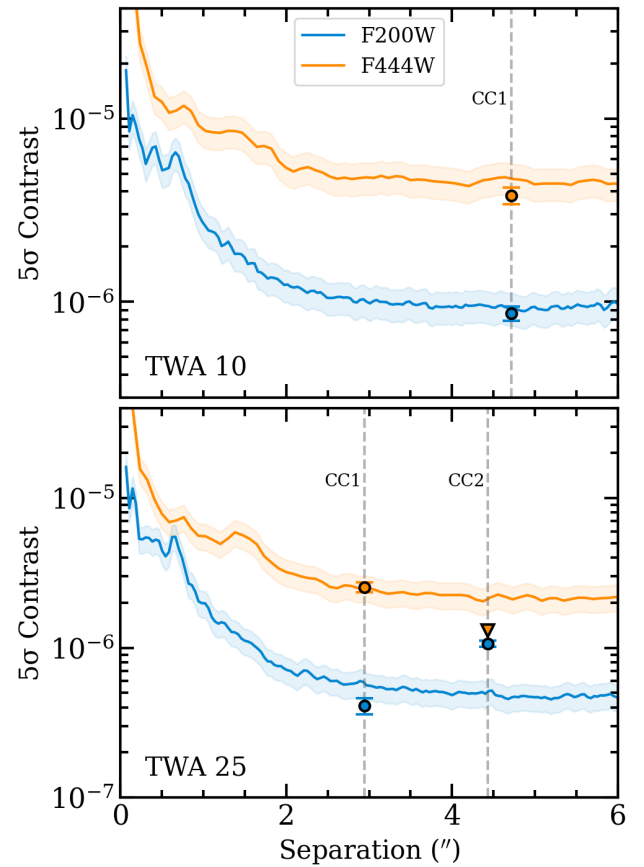


Figure 8. 5σ contrast curves of our disk model subtracted data for TWA 10 (top) and TWA 25 (bottom). The blue lines are the contrasts for the F200W observations, while the orange lines are the contrasts for the F444W observations. The shaded regions represent $\pm 1\sigma$ contrasts. The blue data points are the contrasts for potential candidates observed in F200W, while the orange data points are the contrasts for potential candidates observed in the F444W. This includes one candidate in the TWA 10 system and two potential candidates in the TWA 25 system.

radius (~ 149.6 and ~ 57.5 au for TWA 10 and 25, respectively), stellar mass, and system age, we estimate the maximum semimajor axis and minimum mass for potential planets in each system. These estimations are based on the assumption that a single planet is actively sculpting its disk near the disk inner edge and is responsible for clearing the material interior to the disk.

We use the code `SculptingPlanet`, a model that is used in T. D. Pearce et al. (2022) and builds upon T. D. Pearce & M. C. Wyatt (2014). We assume that the planet has a circular orbit, as our observations do not strongly suggest that either debris disk is eccentric. We also assume a 10% error on the disk inner radius; observations at longer wavelengths, such as with ALMA, will be crucial for more robustly estimating the disk inner edge, as larger grains more closely trace the planetesimal belt compared to the micron-sized grains probed at NIR wavelengths. This leads to an estimated maximum semimajor axis and minimum planet mass of 98 ± 9 au and $1.8 \pm 0.4 M_{\text{jup}}$ for TWA 10 and 41 ± 4 au and $1.1 \pm 0.2 M_{\text{jup}}$ for TWA 25, represented by the pink data points in Figure 9. For larger planets, the semimajor axis decreases slightly, as the planet does not have to be as close to the disk to sculpt it. The relationship between planet mass and semimajor

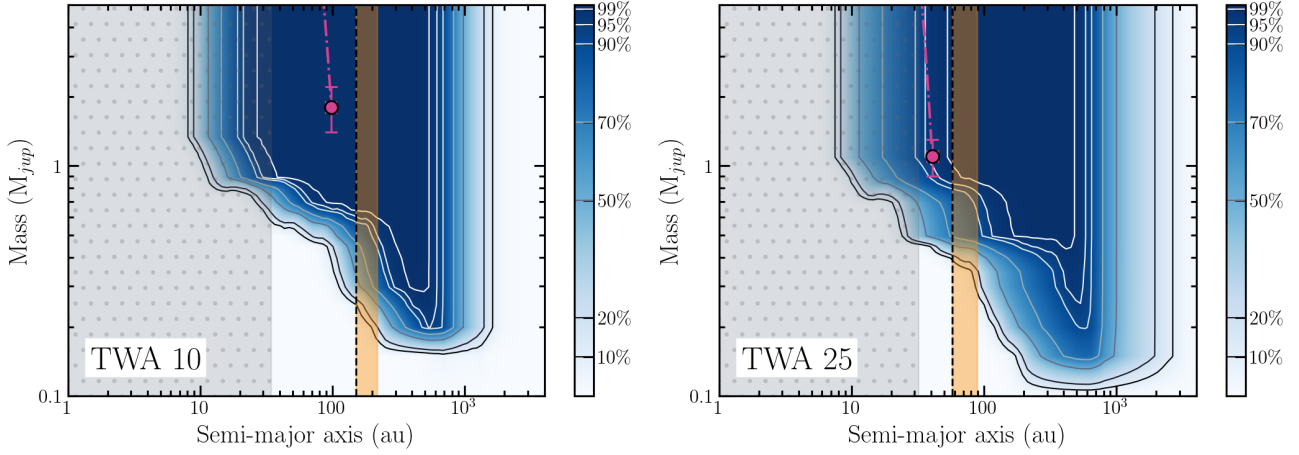


Figure 9. Planet detection probability maps for TWA 10 (left) and TWA 25 (right) as calculated by ExoDMC using the planet-mass sensitivity derived from our 5σ contrast curves. The dashed black line represents the estimated disk inner radius, while the orange shaded region represents the disk extent. The gray shaded region represents the area where the throughput is $\leq 50\%$ (e.g., ≤ 0.6). The pink data point represents the maximum semimajor axis and minimum planet mass expected for a single planet that is actively sculpting the disk inner edge, while the pink dashed-dotted line represents the relationship between planet mass and semimajor axis, both derived using the code from T. D. Pearce et al. (2022).

axis is represented by the pink dashed-dotted lines. Our results will be further discussed in Section 6.3.

6.2. Companion Extractions and Colors

Analyzing our observations within $\sim 5''$, we find one significant point source for TWA 10 and two for TWA 25 that are identifiable by eye above the background noise. One downside to RDI is that objects observed in reference images may be injected into our final reductions. To ensure that none of the point sources were injected by one of the reference images, we use the jackknife test, which removes reference images from our reductions one by one. Doing so, we find that all three point sources pass the jackknife test, meaning that they are indeed present in our observations of TWA 10 and TWA 25. Thus, we now refer to these sources as candidate companions (CC), and they are highlighted in Figure 10. We also note that all the candidates are outside the disk location. Although it is certainly possible for a planet to be orbiting outside the disk (e.g., HD 106906, V. Bailey et al. 2014), these candidates are not related to the planets predicted in Section 6.1 near the disk inner edge.

To estimate their likelihood of being planetary in nature, we first use Winnie to measure their magnitude in each filter. This is done using a point-source model that is forward-modeled and fit to each candidate using `lmfit.minimize` via the same procedure used to find the best-fit disk model. The measured R.A./decl. and apparent magnitudes for each candidate can be found in Table 4 and contrasts are plotted in Figure 8. Because the CC 2 candidate is not present in F444W for the TWA 25 system, we set an upper limit on the magnitude equivalent to a 3σ contrast at that separation. The measured apparent magnitudes are then used to calculate the F200W – F444W color, which we compare to colors measured for galaxies as part of the JADES survey (D. J. Eisenstein et al. 2026), as well as stellar models from the TRILEGAL code (TRIdimensional model of the GALaxy; L. Girardi et al. 2005). Additionally, we compare to the BEX evolutionary model (assuming a system age of 10 Myr) via a color-magnitude diagram (CMD). The resulting CMD is shown in Figure 11.

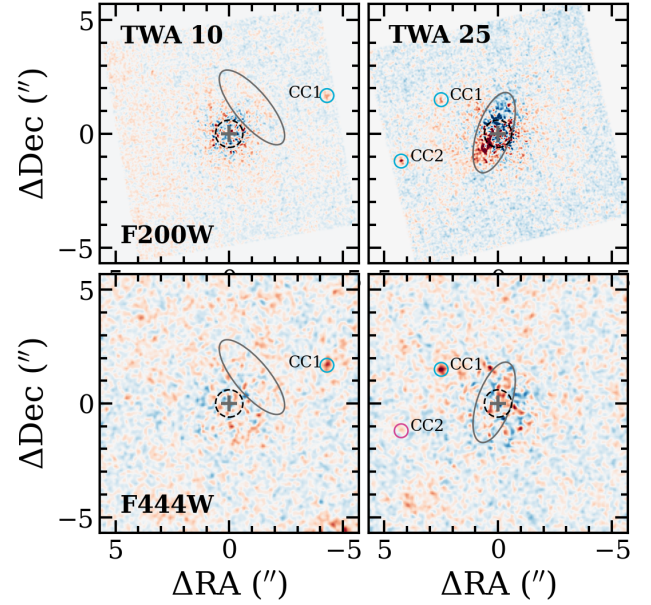


Figure 10. Model subtracted observations for TWA 10 (left) and TWA 25 (right), where the F200W data for both systems is shown in the top row and F444W data is shown in the bottom row. The blue circles represent identified candidates that are present in that specific filter, while pink circles represent identified candidates that are only observed in the opposite filter. Each candidate companion is labeled by CC 1–CC 2. The black dashed circles represent a radius of $0.6''$ where the throughput reaches 50%, while the solid gray ellipses highlight the location of the disk.

We find that all three candidates across both systems are likely too blue in color to be potential planet companions, especially CC 2, which only has an upper limit on color. Additionally, all three candidates lie within the same parameter space as the JADES galaxies and TRILEGAL stars, further suggesting that they are likely background objects. Although we do not yet fully understand the full extent of possible colors and evolutionary tracks for the demographic of low-mass, wide-orbit planets, we do not further consider these three sources as planet candidates in this work.

Table 4Results from Candidate Model Fitting with *spaceKLIP* Including the Measured R.A. and Decl., as well as Apparent Magnitudes in Both Filters

Name	Candidate	R.A. (arcsec)	Decl. (arcsec)	F200W (App. Mag)	F444W (App. Mag)
TWA 10	CC 1	-4.32 ± 0.01	1.70 ± 0.01	23.37 ± 0.10	21.46 ± 0.11
TWA 25	CC 1	2.51 ± 0.01	1.50 ± 0.01	23.26 ± 0.14	21.11 ± 0.08
...	CC 2	4.26 ± 0.01	-1.19 ± 0.01	22.23 ± 0.05	>21.83

Note. Here, a negative R.A. is to the right of the star and a positive R.A. is to the left of the star.

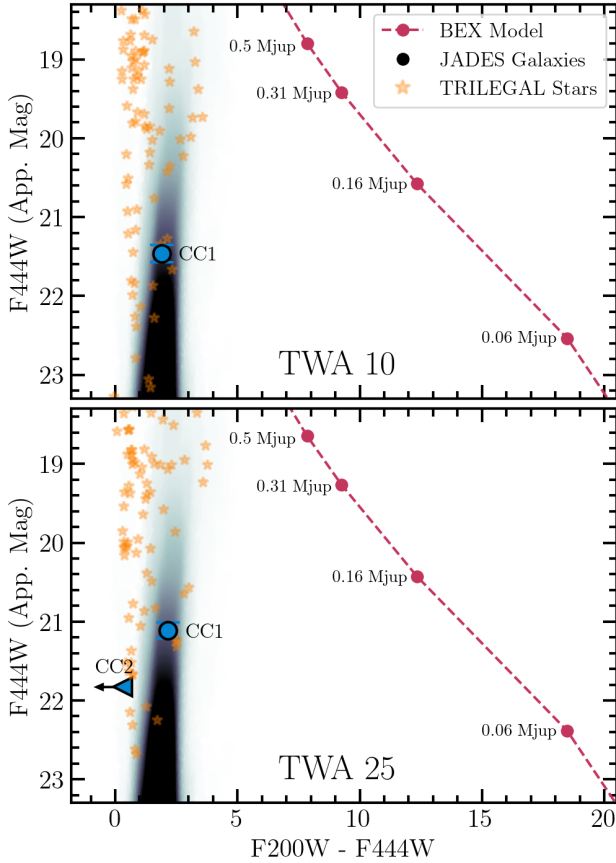


Figure 11. F200W – F444W colors vs. F444W apparent magnitude for TWA 10 (top) and TWA 25 (bottom). The black/gray data points represent the sample of galaxies measured as part of the JADES survey, while the orange stars represent the model stars from TRILEGAL. The blue data points represent the candidate companions in the TWA 10 and TWA 25 systems, which we compare to the BEX evolutionary model (pink dashed line).

6.3. Implications of Exoplanet Nondetections

In the previous two sections, we rule out the three sources found in our observations within $5''$ to be exoplanet companions, as well as infer the possible planet semimajor axis and masses based on the disk inner radius. Given these analyses, we can say something about the implications of both the lack of detection of potential planet candidates in our observations and where we expected to find such candidates based on the debris disk morphologies.

For both the TWA 10 and 25 systems, the predicted planets have a significant probability of being detected in the F444W filter based on the disk inner edge, stellar mass, and age. While the probability is slightly lower for TWA 25 ($\sim 90\%$ – 95% depending on planet mass) given the higher inclination and smaller disk extent, the predicted planet based on the TWA 10 disk would have a 99% probability of being detected. Even in

the case of a larger planet mass and thus decreased required semimajor axis, the planet would still have a 99% probability of being detected in the F444W. Additionally, we create the same detection probability maps using the F200W planet-mass sensitivity curves for comparison, which shows that a planet of mass $\gtrsim 2.5 M_{\text{jup}}$ would also have a 99% probability of being detected in either system. In summary, we should have detected a planet in both the TWA 10 and 25 systems, assuming a single planet sculpting the disk inner edge. In the case of a multi-Jupiter mass planet, we would have also likely detected it in the F200W filter.

In the case of TWA 25, our analysis suggests that there is a $\lesssim 10\%$ chance that a disk sculpting planet of $\lesssim 2.5 M_{\text{jup}}$ (again, planets with masses $\gtrsim 2.5 M_{\text{jup}}$ would have been detected in the F200W filter) may have eluded detection given its small semimajor axis. TWA 10 presents an even more perplexing situation. Regardless of planet mass, if a single planet is responsible for sculpting the disk, we should have detected it with at least 5σ significance. However, we do not detect any significant point sources in either filter outside of CC 1, which we have already determined is most likely a background object. This conundrum raises the question as to why we are not detecting any planet companions in either system when their debris disks suggest that there should be one.

There are several potential scenarios that could explain the lack of a planet detection in the TWA 10 and TWA 25 systems.

1. The atmospheric evolutionary models used overpredict the expected planet-mass sensitivity, given the lack of observed sub-Jupiter mass planets that help inform such models. For example, water clouds, disequilibrium chemistry, and increased metallicity can cause suppression in the F444W band (e.g., R. Bowens-Rubin et al. 2025), which is not accounted for in the E. F. Linder et al. (2019) models. A similar conclusion was reached in the case of TWA 7 b, where the same BEX models predicted a planet mass closer to $0.06 M_{\text{jup}}$ (K. A. Crofts et al. 2025). However, combined photometry of NIRCам and MIRI observations suggested a warm, Saturn-mass planet with a disequilibrium chemistry and above solar metallicity atmosphere (K. A. Crofts et al. 2025; A.-M. Lagrange et al. 2025). Therefore, it is possible that the atmospheric chemistry of sub-Jupiter mass planets present in these systems differ from the chemistry assumed in the BEX models, making them more difficult to detect than expected. A detailed investigation regarding the effects of clouds and chemistry on the estimated sensitivity from the NIRCам surveys (including GO 4050) are currently being conducted in a separate effort (G. M. Strampelli 2026, in preparation).
2. Multiple smaller planets are sculpting the debris disks rather than one single larger planet, making them out of

reach of direct detection, even in the case of the BEX evolutionary model. This scenario is also explored in T. D. Pearce et al. (2022), assuming multiple planets of equal mass, where they found that the planets in the multiplanet scenario were on average 40 times less massive than a single planet. This scenario is also supported for the TWA 25 system, where the sharp inner edge of the disk ($\alpha_{\text{in}} = 11.94 \pm 5.37$) suggests sculpting by multiple smaller planets, as more massive planets excite dust eccentricities and cause the inner edge to smooth out (T. D. Pearce et al. 2024).

3. Planets may have started closer to the disk early on and then migrated inwards over time, beyond what we can detect given the IWA of NIRCcam. This hypothesis has been suggested in recent studies, where T. D. Pearce et al. (2024) found that, on average, the inner edges of debris disks were too steep to be caused solely by collisional activity, but not steep enough to be solely due to planet sculpting. In summary, the process of inward migration would cause debris caught in resonance by the planet to become excited on increasingly higher eccentricities (M. C. Wyatt 2003; R. Reche et al. 2008; M. F. Friebe et al. 2022), resulting in a more shallow inner edge profile.
4. The location of the debris disk inner edge is unrelated to planets, but is rather set by processes during the protoplanetary disk phase. For example, planetesimals may naturally form in distinct radial zones via streaming instability (D. Carrera et al. 2017) or snow lines (e.g., S. Ida & T. Guillot 2016; J. Drazkowska & Y. Alibert 2017; D. Schoonenberg & C. W. Ormel 2017; D. Schoonenberg et al. 2018; A. Izidoro et al. 2022; A. Morbidelli et al. 2022). Other mechanisms can also stir planetesimals in the disk to produce a collisional cascade, such as self-stirring (if the disk is massive enough; S. J. Kenyon & B. C. Bromley 2001) or stellar flybys (S. Ida et al. 2000; S. J. Kenyon & B. C. Bromley 2002).

7. Summary

We presented new scattered-light observations of the M dwarf debris disks around TWA 10 and TWA 25 with NIRCcam, taken in the F200W and F444W filters. While the TWA 25 debris disk has been previously imaged with HST and SPHERE, NIRCcam has provided the most sensitive observations of the disk to date, probing the disk substructure on a much deeper level. Additionally, our observations have led to the discovery of the TWA 10 debris disk, providing the opportunity to characterize the disk for the very first time.

Through modeling and analysis of each disk, we find the following results:

1. The TWA 10 disk is by far the largest M dwarf debris disk detected to date, with a radius of ~ 191 au. This finding suggests that M dwarf debris disks may have a larger range of radii than previously thought, bringing into question the possible evolutionary tracks for these disks and the relationship between disk radius and stellar luminosity.
2. We are able to probe the TWA 25 disk morphology with higher sensitivity than other scattered-light observations, allowing for greater characterization of the disk. We find

that the disk radius is smaller than previous estimations, where we derive a radius of ~ 63.8 au compared to ~ 76 – 78 au. Additionally, we find that the disk has a very steep inner dust density slope of 11.94 ± 5.37 , suggesting that the disk inner radius may be actively sculpted by planets. Measuring the surface brightness reveals a small but significant brightness asymmetry between the SE and NW sides of the disk in the F200W, while the disk appears axisymmetric in F444W.

3. Analysis of the potential disk SED for both systems, based on our NIRCcam data and limits set by MIPS and PACS, suggests that the IR excess was not robustly detected in the MIR–FIR likely due to low luminosity fractions. However, future observations of the disk emission, such as with ALMA, will be required to place more robust constraints the disk SEDs.

In addition to analyzing the disk morphology, we also use our observations to search for potential planetary candidates. We find one potential candidate in the TWA 10 system and two in the TWA 25 system within $5''$. Color analysis of these candidates reveal that they are likely background objects as they are too blue and overlap with both galaxies observed in the JADES survey and stellar models from the TRILEGAL code.

Finally, we create planet detection probability maps using the measured 5σ contrast curves and ATMO/BEX evolutionary models and calculate the expected minimum planet mass and maximum semimajor axis assuming a single planet actively sculpting the disk inner edge. We find that such a planet should have been detected in our observations with a high probability. Thus, the nondetection of any planet candidates within the disk inner radius suggests that either the planet atmospheric evolutionary models assumed are incorrect, multiple smaller planets are sculpting the disk rather than a single giant planet, planets are migrating inwards over time, the disk inner edge is not related to planets, or a combination of two or more.

In summary, the NIRCcam instrument is an amazing tool for the discovery and analysis of M dwarf debris disks, both due to its wavelength range and sensitivity. This has led to new insights that will be important to better understanding the architectures and evolution of such systems compared to those around other spectral types.

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Facility: JWST.

Software: spaceKLIP (J. Kammerer et al. 2022; A. L. Carter et al. 2023, <https://spaceklip.readthedocs.io/en/latest/index.html>), Winnie (K. Lawson et al. 2022, <https://github.com/kdlawson/Winnie>), lmfit (M. Newville et al. 2025, <https://lmfit.github.io/lmfit-py/>), matplotlib (J. D. Hunter 2007), iPython (F. Perez & B. E. Granger 2007), Astropy (Astropy Collaboration et al. 2013, 2018, 2022), NumPy (C. R. Harris et al. 2020; <https://numpy.org>), SciPy (P. Virtanen et al. 2020; <http://www.scipy.org/>).

Appendix A SED Photometry

In Section 4.3, we compiled photometry to build the SED for both TWA 10 and TWA 25, in order to understand why no significant IR excess has been detected for these systems. The photometry for each filter, as well as the appropriate references, can be found below in Table 5.

Table 5

Photometry for the TWA 10 and TWA 25 SEDs Plotted in Figure 5

Name	Filter	Flux (Jy)	Reference
TWA 10	BAPASS	$5.9\text{e-}3 \pm 2.3\text{e-}4$	(1)
	GAPASS	$1.1\text{e-}2 \pm 4.6\text{e-}4$	(1)
	GAIA.BP	$1.5\text{e-}2 \pm 2.9\text{e-}4$	(2)
	VAPASS	$2.2\text{e-}5 \pm 9.9\text{e-}4$	(1)
	RAPASS	$3.8\text{e-}2 \pm 1.8\text{e-}3$	(1)
	GAIA.G	$5.2\text{e-}2 \pm 5\text{e-}4$	(2)
	IAPASS	$1.3\text{e-}1 \pm 8.0\text{e-}3$	(1)
	GAIA.RP	$1.2\text{e-}1 \pm 1.5\text{e-}3$	(2)
	2MASS J	$3.5\text{e-}1 \pm 8.3\text{e-}3$	(3)
	2MASS H	$4.3\text{e-}1 \pm 1.8\text{e-}2$	(3)
	2MASS Ks	$3.5\text{e-}2 \pm 9.9\text{e-}3$	(3)
	WISE3	$1.7\text{e-}1 \pm 4.2\text{e-}3$	(4)
	WISE4	$1.1\text{e-}1 \pm 2.9\text{e-}3$	(4)
	WISE12	$2.2\text{e-}2 \pm 9.6\text{e-}4$	(5)
	WISE22	$6.7\text{e-}3 \pm 8.2\text{e-}4$	(5)
	MIPS24	$5.8\text{e-}3 \pm 3.0\text{e-}5$	(6)
	MIPS70	$<8.0\text{e-}3$	(7)
	PACS70	$<4.1\text{e-}3$	(8)
	PACS160	$<1.2\text{e-}2$	(8)
	PACS250	$<2.70\text{e-}2$	(11)
	PACS350	$<2.70\text{e-}2$	(11)
	PACS500	$<2.25\text{e-}2$	(11)

Table 5
(Continued)

Name	Filter	Flux (Jy)	Reference
TWA 25	BT	$1.4\text{e-}2 \pm 3.7\text{e-}3$	(9)
	BAPASS	$3.7\text{e-}2 \pm 4.7\text{e-}3$	(1)
	GAPASS	$6.3\text{e-}2 \pm 8.4\text{e-}3$	(1)
	GAIA.BP	$8.4\text{e-}2 \pm 1.2\text{e-}3$	(2)
	VT	$8.4\text{e-}2 \pm 7.3\text{e-}3$	(9)
	VAPASS	$1.3\text{e-}1 \pm 9.9\text{e-}3$	(1)
	RAPASS	$2.1\text{e-}1 \pm 3.2\text{e-}2$	(1)
	GAIA.G	$2.0\text{e-}1 \pm 1.9\text{e-}3$	(2)
	IAPASS	$3.5\text{e-}1 \pm 6.4\text{e-}3$	(1)
	GAIA.RP	$3.8\text{e-}1 \pm 4.4\text{e-}3$	(2)
	2MASS J	$8.4\text{e-}1 \pm 2.7\text{e-}2$	(3)
	2MASS H	$1.1\text{e}0 \pm 4.2\text{e-}2$	(3)
	2MASS Ks	$7.9\text{e-}1 \pm 1.6\text{e-}2$	(3)
	WISE3	$3.6\text{e-}1 \pm 9.1\text{e-}3$	(4)
	WISE4	$2.2\text{e-}1 \pm 7.0\text{e-}3$	(4)
	AKARI9	$1.1\text{e-}1 \pm 1.6\text{e-}2$	(10)
	WISE12	$4.2\text{e-}2 \pm 1.9\text{e-}3$	(5)
	WISE22	$1.3\text{e-}2 \pm 1.1\text{e-}3$	(5)
	MIPS24	$1.1\text{e-}2 \pm 2.3\text{e-}4$	(6)
	MIPS 70	$5.9\text{e-}3 \pm 4.0\text{e-}3$	(6)
	PACS70	$<5.7\text{e-}3$	(8)
	PACS100	$<6.3\text{e-}3$	(8)
	PACS160	$<1.2\text{e-}2$	(8)

Note. References: (1) A. A. Henden et al. (2016), (2) Gaia Collaboration (2018), (3) M. F. Skrutskie et al. (2006), (4) F. Marocco et al. (2021), (5) E. L. Wright et al. (2010), (6) SSC & IRSA (2021), (7) F. J. Low et al. (2005), (8) P. Riviere-Marichalar et al. (2013), (9) E. Høg et al. (2000), (10) D. Ishihara et al. (2010), (11) L. A. Cieza et al. (2013).

Appendix B Additional TWA 25 F444W Objects

Although the TWA 25 system only has two candidates within both the F200W and F444W FOVs, there are five objects in the F444W data located outside the F200W FOV ($\gtrsim 5''$). These objects are labeled B1–B5 in Figure 12. While B2–B5 appear point-source like, B1 is clearly extended and, therefore, is very likely a background galaxy. For B2–B5, we cannot make any claims on whether or not these objects are bound companions or background objects, as we cannot measure their colors, although it is less likely that these are bound companions given their distance from the star. Proper motion follow-up would be required to more confidently determine that these are indeed background objects. Similar to the three candidates in Section 6.2, we use Winnie and `lmfit.minimize` to fit each point source with a forward model in order to estimate their RA/Dec and apparent magnitude. The results from this forward modeling can be found in Table 6.

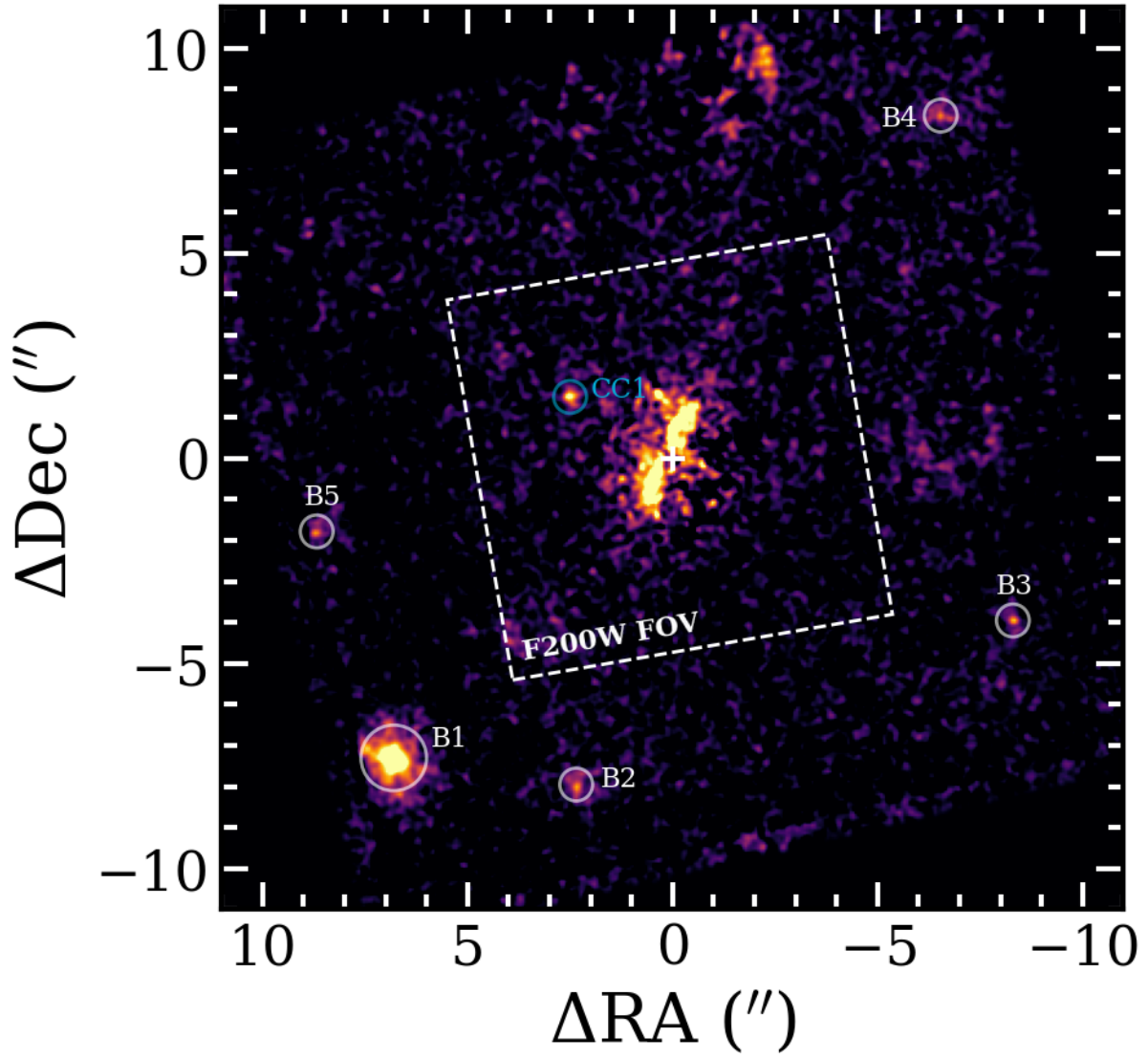


Figure 12. TWA 25 F444W data zoomed out to show the additional sources, which are circled in white and labeled B1–B5. The white dashed box represents the FOV of the F200W observations. Additionally, CC 1 is circled in blue.

Table 6

Results from Candidate Model Fitting with *spaceKLIP* for the TWA 25 Objects Outside of the F200W FOV Including the Measured R.A. and Decl., as well as Apparent Magnitude in the F444W Filter

Candidate	R.A. (arcsec)	Decl. (arcsec)	F444W (App. Mag)
B1	6.79 ± 0.01	-7.37 ± 0.01	20.00 ± 0.08
B2	2.33 ± 0.01	-8.02 ± 0.01	21.28 ± 0.09
B3	-8.31 ± 0.01	-3.97 ± 0.01	21.31 ± 0.09
B4	-6.80 ± 0.02	8.25 ± 0.01	21.53 ± 0.13
B5	8.69 ± 0.01	-1.84 ± 0.01	21.54 ± 0.12

Note. A negative R.A. is to the right of the star and a positive R.A. is to the left of the star.

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